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Matthew **Crump**

Professor of Psychology

Positions Held

2024-present	Chairperson, Department of Psychology, Brooklyn College of CUNY
2022-present	Full Professor, Department of Psychology, Brooklyn College of CUNY
2016-2022	Associate Professor, Department of Psychology, Brooklyn College of CUNY
2011-2016	Assistant Professor, Department of Psychology, Brooklyn College of CUNY
2011-present	Doctoral Faculty, Psychology, Graduate Center of CUNY
2007-2011	Postdoctoral Researcher, Department of Psychology, Vanderbilt University

Education

2007	Ph.D. Psychology, McMaster University, Hamilton, ON, Canada
2002	B.Sc. (Hon.) Psychology, University of Lethbridge, Lethbridge, AB, Canada

Publications

(* = Postdoctoral Associate, ** = Doctoral Student, *** = Undergraduate Student)

1 Johns, B. T., Jamieson, R. K., **Crump, M. J. C.**, & Jones, M. N. (2026). Instance memory models as a general computational framework for exploring language processing: Bringing the lexicon to life. R. *Linguistics Vanguard*, *12*(s1), 163-172. https://doi.org/10.1515/lingvan-2024-0201

2 Jamieson, R. K., & **Crump, M. J. C.** (2025). An instance model of associative inference. R. *The American Journal of Psychology*, *138*(2), 231-247. https://scholarlypublishingcollective.org/uip/ajp/article-abstract/138/2/231/404898/An-Instance-Model-of-Associative-Inference

3 Schneider, D. W., **Crump, M. J. C.**, & Verbruggen, F. (2025). Preface. *The American Journal of Psychology*, *138*(2). Preface to the Gordon D. Logan Festschrift.

4 *Behmer, L. P., Jr., **Crump, M. J. C.**, & Jantzen, K. J. (2023). Motor-evoked potentials for early individual elements of an action sequence during planning reflect parallel activation processes. R. *Motor Control*. Published online ahead of print.

5 **Brosowsky, N. P., & **Crump, M. J. C.** (2021). Contextual recruitment of selective attention can be updated via changes in task relevance. R. *Canadian Journal of Experimental Psychology*, *75*, 19-34. https://doi.org/10.1037/cep0000221

6 **Crump, M. J. C.** (2021). *Instances of cognition: Questions, methods, findings, explanations, applications, and implications*. Undergraduate textbook. Open educational resource. https://www.crumplab.com/cognition/textbook/

7 **Crump, M. J. C.** (2021). *Reproducible statistics for psychologists with R: Lab tutorials*. Graduate textbook. Open educational resource. https://www.crumplab.com/rstatsforpsych

8 Vuorre, M., & **Crump, M. J. C.** (2021). Sharing and organizing research products as R packages. R. *Behavior Research Methods*, *53*, 792-802. https://doi.org/10.3758/s13428-020-01436-x

9 **Crump, M. J. C.**, Jamieson, R. K., Johns, B. T., & Jones, M. N. (2020). Controlling the retrieval of general vs specific semantic knowledge in the instance theory of semantic memory. R. In S. Denison, M. Mack, Y. Xu, & B. C. Armstrong (Eds.), *Proceedings of the 42nd Annual Conference of the Cognitive Science Society* (pp. 3261-3267). Cognitive Science Society.

10 Johns, B. T., Jamieson, R. K., **Crump, M. J. C.**, Jones, M. N., & Mewhort, D. J. K. (2020). Production without rules: Using an instance memory model to exploit structure in natural language. R. *Journal of Memory and Language*, *115*, 104165. https://doi.org/10.1016/j.jml.2020.104165

11 Aujla, H., **Crump, M. J. C.**, Cook, M. T., & Jamieson, R. K. (2019). The Semantic Librarian: A search engine built from vector-space models of semantics. R. *Behavior Research Methods*, *51*, 2405-2418. https://doi.org/10/gf6cmv

12 Braem, S., Bugg, J. M., Schmidt, J. R., **Crump, M. J. C.**, Weissman, D. H., Notebaert, W., & Egner, T. (2019). Measuring adaptive control in conflict tasks. R. *Trends in Cognitive Sciences*, *23*, 769-783. https://doi.org/10/gf48x7

13 **Crump, M. J. C.** (2019). Portfolio and prosper. Invited. *Nature Human Behaviour*, *3*, 1008. https://doi.org/10.1038/s41562-019-0733-0

14 **Crump, M. J. C.**, ***Lai, W., & **Brosowsky, N. P. (2019). Instance theory predicts information theory: Episodic uncertainty as a determinant of keystroke dynamics. R. *Canadian Journal of Experimental Psychology*, *73*, 203-215.

15 *Behmer, L. P., Jr., Jantzen, K. J., Martinez, S., Walls, R., Amir-Brownstein, E., Jaye, A., Leytze, M., Lucier, K., & **Crump, M. J. C.** (2018). Parallel regulation of past, present, and future actions during sequencing. R. *Journal of Experimental Psychology: Human Perception and Performance*, *44*(8), 1147-1152. https://doi.org/10.1037/xhp0000507

16 **Brosowsky, N. P., & **Crump, M. J. C.** (2018). Memory-guided selective attention: Experiences with conflict have long-lasting effects on cognitive control. R. *Journal of Experimental Psychology: General*, *147*(8), 1134-1153.

17 **Crump, M. J. C.**, Krishnan, A., Volz, S., Chavarga, A., & Suzuki, J. (2018). *Answering questions with data: The lab manual*. Undergraduate textbook. Open educational resource. https://crumplab.github.io/statisticsLab/

18 **Crump, M. J. C.**, Milliken, B., Leboe-McGowan, J. P., Leboe-McGowan, L. C., & Gao, X. (2018). Context-dependent control of attention capture: Evidence from proportion congruent effects. R. *Canadian Journal of Experimental Psychology*, *72*, 91-104. https://doi.org/10.1037/cep0000145

19 **Crump, M. J. C.**, Navarro, D., & Suzuki, J. (2018). *Answering questions with data: Introductory statistics for psychology students*. Undergraduate textbook. Open educational resource. https://crumplab.github.io/statistics/

20 *Behmer, L. P., & **Crump, M. J. C.** (2017). The dynamic range of response set activation during action sequencing. R. *Journal of Experimental Psychology: Human Perception and Performance*, *43*(3), 537-554.

21 *Behmer, L. P., & **Crump, M. J. C.** (2017). Crunching big data with finger tips: How typists tune their performance toward the statistics of natural language. R. In M. N. Jones (Ed.), *Big Data in Cognitive Science* (pp. 319-341). Routledge.

22 *Behmer, L. P., & **Crump, M. J. C.** (2017). Spatial knowledge during skilled action sequencing: Hierarchical versus non-hierarchical representations. R. *Attention, Perception & Psychophysics*, *79*, 2435-2448. https://doi.org/10.3758/s13414-017-1389-3

23 **Crump, M. J. C.** (2017). *Programming for psychologists: Data creation and analysis*. Undergraduate textbook. Open educational resource. https://crumplab.github.io/programmingforpsych/

24 **Crump, M. J. C.**, **Brosowsky, N. P., & Milliken, B. (2017). Reproducing the location-based context-specific proportion congruent effect for frequency unbiased items: A reply to Hutcheon & Spieler (2016). R. *The Quarterly Journal of Experimental Psychology*, *70*, 1792-1807.

25 **Crump, M. J. C.**, Price, P. C., Jhangiani, R., Chiang, I.-C. A., & Leighton, D. C. (2017). *Research methods in psychology*. Undergraduate textbook. Open educational resource. https://crumplab.github.io/ResearchMethods/

26 Zumsteg, J. W., **Crump, M. J. C.**, Logan, G. D., Weikert, D. R., & Lee, D. H. (2017). The effect of carpal tunnel release on typing performance. R. *The Journal of Hand Surgery*, *42*, 17-23. https://doi.org/10.1016/j.jhsa.2016.10.005

27 **Brosowsky, N. P., & **Crump, M. J. C.** (2016). Context-specific attentional sampling: Intentional control as a pre-requisite for contextual control. R. *Consciousness and Cognition*, *44*, 146-160. https://doi.org/10.1016/j.concog.2016.07.001

28 **Crump, M. J. C.** (2016). Learning to selectively attend from context-specific attentional histories: A demonstration and some constraints. R. *Canadian Journal of Experimental Psychology*, *70*, 59-77.

29 Curtis, E. T., Chubala, C. M., Spear, J., Jamieson, R. K., Hockley, W. E., & **Crump, M. J. C.** (2016). False recognition of instruction-set lures. R. *Memory*, *24*, 32-43.

30 Johns, B. T., Jamieson, R. K., **Crump, M. J. C.**, Jones, M. N., & Mewhort, D. J. K. (2016). The combinatorial power of experience. R. In *Proceedings of the 38th Annual Conference of the Cognitive Science Society*. Cognitive Science Society.

31 **Crump, M. J. C.**, & Logan, G. D. (2013). Prevention and correction in post-error performance: An ounce of prevention, a pound of cure. R. *Journal of Experimental Psychology: General*, *142*, 692-709.

32 **Crump, M. J. C.**, McDonnell, J. V., & Gureckis, T. M. (2013). Evaluating Amazon's mechanical Turk as a tool for experimental behavioral research. R. *PLoS ONE*, *8*(3), e57410.
33 Yamaguchi, M., **Crump, M. J. C.**, & Logan, G. D. (2013). Speed accuracy-tradeoff in skilled typewriting: Decomposing the contributions of hierarchical control loops. R. *Journal of Experimental Psychology: Human Perception and Performance*, *39*(3), 678-699.
34 Bugg, J. M., & **Crump, M. J. C.** (2012). In support of a distinction between voluntary and stimulus-driven control: A review of the literature on proportion congruent effects. R. *Frontiers in Psychology*, *3*(367), 1-16.
35 **Crump, M. J. C.** (2012). Review of Guitar Zero: The new musician and the science of learning.
36 **Crump, M. J. C.**, Logan, G. D., & Kimbrough, J. (2012). Keeping an eye on guitar skill: Visual representations of guitar chords. R. *Music Perception: An Interdisciplinary Journal*, *30*(1), 37-47.
37 Jamieson, R. K., **Crump, M. J. C.**, & Hannah, S. D. (2012). An instance theory of associative learning. R. *Learning & Behavior*, *40*(1), 61-82.
38 Logan, G. D., & **Crump, M. J. C.** (2011). Hierarchical control of cognition and action: The case for skilled typewriting. R. In B. H. Ross (Ed.), *Psychology of Learning and Motivation* (Vol. 54, pp. 1-27). Elsevier.
39 **Crump, M. J. C.**, & Logan, G. D. (2010). Warning, this keyboard will de-construct: The role of the keyboard in skilled typewriting. R. *Psychonomic Bulletin & Review*, *17*(3), 394-399.
40 **Crump, M. J. C.**, & Logan, G. D. (2010). Contextual control over task-set retrieval. R. *Attention, Perception & Psychophysics*, *72*(8), 2047-2053.
41 **Crump, M. J. C.**, & Logan, G. D. (2010). Hierarchical control and skilled typing: Evidence for word level control over the execution of individual keystrokes. R. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *36*(6), 1369-1380.
42 **Crump, M. J. C.**, & Logan, G. D. (2010). Episodic contributions to sequential control: Learning from a typist's touch. R. *Journal of Experimental Psychology: Human Perception and Performance*, *36*(3), 662-672.
43 Jamieson, R. K., Hannah, S. D., & **Crump, M. J. C.** (2010). A memory-based account of retrospective revaluation. R. *Canadian Journal of Experimental Psychology*, *64*(3), 153-164.
44 Liu, X., **Crump, M. J. C.**, & Logan, G. D. (2010). Do you know where your fingers have been? Explicit knowledge of the spatial layout of the keyboard in skilled typists. R. *Memory & Cognition*, *38*(4), 474-484.
45 Logan, G. D., & **Crump, M. J. C.** (2010). Cognitive illusions of authorship reveal hierarchical error detection in skilled typists. R. *Science*, *330*(6004), 683-686.
46 **Crump, M. J. C.**, & Milliken, B. (2009). The flexibility of context-specific control: Evidence for context-driven generalization of item-specific control settings. R. *The Quarterly Journal of Experimental Psychology*, *62*(8), 1523-1532.
47 Hannah, S. D., **Crump, M. J. C.**, Allan, L. G., & Siegel, S. (2009). Cue-interaction effects in contingency judgments using the streamed-trials procedure. R. *Canadian Journal of Experimental Psychology/Revue canadienne de psychologie expérimentale*, *63*(2), 103-112.
48 Logan, G. D., & **Crump, M. J. C.** (2009). The left hand doesn't know what the right hand is doing: The disruptive effects of attention to the hands in skilled typewriting. R. *Psychological Science*, *20*(10), 1296-1300.
49 Siegel, S., Allan, L. G., Hannah, S. D., & **Crump, M. J. C.** (2009). Applying signal detection theory to contingency assessment. R. *Comparative Cognition & Behavior Reviews*, *4*, 116-134.
50 Allan, L. G., Hannah, S. D., **Crump, M. J. C.**, & Siegel, S. (2008). The psychophysics of contingency assessment. R. *Journal of Experimental Psychology: General*, *137*(2), 226-243.
51 **Crump, M. J. C.**, Milliken, B., & Ansari, I. (2008). Shifting views on the symbolic cueing effect: Cueing attention through previous experience. R. *Psicológica*, *29*(1), 97-114.
52 **Crump, M. J. C.**, Vaquero, J. M. M., & Milliken, B. (2008). Context-specific learning and control: The role of awareness, task-relevance, and relative salience. R. *Consciousness and Cognition*, *17*(1), 22-36.
53 Leboe, J. P., Wong, J., **Crump, M. J. C.**, & Stobbe, K. (2008). Probe-specific proportion task repetition effects on switching costs. R. *Perception & Psychophysics*, *70*(6), 935-945.
54 **Crump, M. J. C.**, Hannah, S. D., Allan, L. G., & Hord, L. K. (2007). Contingency judgments on the fly. R. *The Quarterly Journal of Experimental Psychology*, *60*(6), 753-761.
55 Schmidt, J. R., **Crump, M. J. C.**, Cheesman, J., & Besner, D. (2007). Contingency learning without awareness: Evidence for implicit control. R. *Consciousness and Cognition*, *16*(2), 421-435.
56 **Crump, M. J. C.**, Gong, Z., & Milliken, B. (2006). The context-specific proportion congruent effect: Location as a contextual cue. R. *Psychonomic Bulletin & Review*, *13*(2), 316-321.

Preprints and Works in Progress

1 **Brosowsky, N. P., Parshina, O., Locicero, A., & **Crump, M. J. C.** (2021). Teaching undergraduate students to read empirical articles: An evaluation and revision of the QALMRI method. PsyArXiv. https://doi.org/10.31234/osf.io/p39sc

2 Piech, R. M., **Crump, M. J. C.**, & Zald, D. H. (2021). The negative contingency illusion: A cognitive bias leading to misjudgement of protection. Research Square. https://doi.org/10.21203/rs.3.rs-671507/v1

Invited Talks

1 **Crump, M. J. C.** (2022). Instance theory as a domain-general intuition pump for cognition. CCP colloquium @ The Graduate Center of CUNY.

2 **Crump, M. J. C.** (2019). Betrayed by your fingertips: Keystroke dynamics in typing as cognitive fingerprints. Brain and Cognitive Sciences Seminar @ University of Manitoba, Winnipeg, Manitoba, Canada.

3 **Crump, M. J. C.** (2018). Betrayed by your fingertips: Keystroke dynamics in typing as cognitive fingerprints. ConCats seminar @ NYU, New York, NY.

4 **Crump, M. J. C.** (2018). Creating Interactive Open Educational Resources with Hypothesis. Faculty Day Conference @ Brooklyn College, Brooklyn, NY.

5 **Crump, M. J. C.** (2018). Coordinating cognition at the fingertips: Insights from investigations into skilled typing ability. Department of Psychology, City Tech, Brooklyn, NY.

6 **Crump, M. J. C.** (2017). Coordinating cognition at the fingertips: Insights from investigations into skilled typing ability. Colloquium series, Department of Psychology, Bard College, Annandale-On-Hudson, NY.

7 **Crump, M. J. C.** (2016). The ways prior experience guides cognitive control: Proactive, Reactive, and retrieval-driven control. Cognition and Perception Seminar, Department of Psychology, Neuroscience, and Behavior, McMaster University, ON, Canada.

8 **Crump, M. J. C.**, & Gureckis, T. M. (2016). Uses and challenges of Mechanical Turk. Invited Workshop at the Association for Psychological Science, Chicago, IL.

9 **Crump, M. J. C.**, & Gureckis, T. M. (2015). Uses and challenges of Mechanical Turk. Invited Workshop at the Association for Psychological Science, New York, NY.

10 **Crump, M. J. C.** (2014). Coordinated Cognition: Insights into hierarchical control processes from the Psychology of skilled typewriting. Colloquium series, Department of Psychology, University of Manitoba, MB, CA.

11 **Crump, M. J. C.** (2014). Associative learning of attentional control strategies: Evidence from the proportion congruent literature. Colloquium series, Department of Psychology, New York University, New York, NY.

12 **Crump, M. J. C.** (2013). Coordinated Cognition: Insights into hierarchical control processes from the Psychology of skilled typewriting. Colloquium series, Department of Psychology, McMaster University, ON, CA.

13 **Crump, M. J. C.** (2012). Coordinated Cognition: Insights into hierarchical control processes from the Psychology of skilled typewriting. Cognitive Lunch Seminar, Department of Psychology, Columbia University, New York, NY.

14 **Crump, M. J. C.** (2012). Coordinated Cognition: Insights into hierarchical control processes from the Psychology of skilled typewriting. Cognitive Seminar, Department of Psychology, University of Granada, Spain.

15 **Crump, M. J. C.** (2010). Coordinating learning, memory, and attention during performance: Insights from the psychology of skilled typewriting. Invited Job talk. Psychology Department at Brooklyn College of CUNY.

Abstracts and Papers Contributed at Professional Meetings

1 **Crump, M. J. C.** (2023). Extending global-matching models to pictures and the Vokey effect. Canadian Society for Brain and Behavioral Sciences. Guelph.

2 **Shives, D. W., ***Ihejirika, P., & **Crump, M. J. C.** (2022). The role of stimulus duration in directed forgetting for natural scenes. Psychonomics. Boston.

3 Jamieson, R. K., & **Crump, M. J. C.** (2021). An instance-based theory of implicit inference. Canadian Society for Brain, Behavior and Cognitive Science, Montreal.

4 **Crump, M. J. C.**, & Vuorre, M. (2020). Creating, curating, and communicating reproducible research with R packages: Vertical as a full-stack example. Society for Computation in Psychology, Online.

5 **Crump, M. J. C.**, Jamieson, R. K., & Aujla, H. (2019). Semantic Librarian for SCIP Abstracts: An R Shiny App and Package for Searching and Visualizing Semantic Spaces. Society for Computation in Psychology, Montreal.

6 Chammany, M. B., Palkowitz, S. R., **Crump, M. J. C.**, & *Behmer, L. P. (2019). Changes in Cognitive Control After High-Demand Trials During Continuous Motor Performance: Slower Rts and Increases in Midline Theta. Psychonomics, Montreal.

7 Chammany, M. B., Palkowitz, S. R., **Crump, M. J. C.**, & *Behmer, L. P. (2019). Increases in Midline and Frontal Theta Reflect the Degree of Cognitive Control Required During Continuous Motor Performance When Explicitly Monitoring Output. Psychonomics, Montreal.

8 Jamieson, R. K., Aujla, H., **Crump, M. J. C.**, & Cook, M. (2018). Using semantic memory theory to build an applied cognitive search engine. Society for Computers in Psychology, New Orleans.

9 **Crump, M. J. C.**, & **Brosowsky, N. B. (2018). Instances and information: An instance theory account of sensitivity to uncertainty in natural text production. Society for Computers in Psychology. New Orleans.

10 **Brosowsky, N. B., & **Crump, M. J. C.** (2018). You should hate this movie! Detecting hidden attitudes of online persuaders. Psychonomics, New Orleans.

11 **Brosowsky, N. B., & **Crump, M. J. C.** (2017). Same Context, Different Control: Flexibly Shifting Between Context-Specific Control Settings. Psychonomics, Vancouver.

12 **Brosowsky, N. B., & **Crump, M. J. C.** (2017). Learning to Selectively Ignore Information. Eastern Psychological Association, Boston.

13 *Behmer, L. P., **Crump, M. J. C.**, Jantzen, K. J., Martinez, S., Walls, R., Amir-Brownstein, E., Jaye, A., Leytze, M., & Lucier, K. (2017). Motor evoked potentials reflect changes in rapid inhibitory control during serial ordering. Cognitive Neuroscience Society, San Francisco.

14 *Behmer, L. P., & **Crump, M. J. C.** (2017). Cognitive Control for Action Sequencing. Eastern Psychological Association, Boston.

15 *Behmer, L. P., & **Crump, M. J. C.** (2016, November). Theta-event related synchrony (ERS) as an index of outer-loop control during typing: Neural evidence supporting the two-loop theory of typing. Psychonomics, Boston, MA.

16 **Brosowsky, N. P., & **Crump, M. J. C.** (2016, November). Long(er)-term item-specific Gratton effects. Psychonomics, Boston, MA.

17 Jamieson, R. K., Chubala, C. M., **Crump, M. J. C.**, & Hannah, S. D. (2016, June). A computational analysis of learning and heuristic inference. Canadian Society for Brain, Behaviour, and Cognitive Science, University of Ottawa, Ottawa, ON, Canada.

18 Johns, B. T., Jamieson, R. K., **Crump, J. C.**, Jones, M. N., & Mewhort, D. J. K. (2016, June). The combinatorial power of experience. Canadian Society for Brain, Behaviour, and Cognitive Science, University of Ottawa, Ottawa, ON, Canada.

19 *Behmer, L. P., & **Crump, M. J. C.** (2016). Try not to think about what you're doing: Event-related desynchronization (ERD) in the SMA and event-related synchronization (ERS) in the frontal lobe reveals a dissociation of hierarchically controlled outer and inner loop processes during typing. Cognitive Neuroscience Society. New York, NY.

20 **Brosowsky, N. P., & **Crump, M. J. C.** (2016). How typists talk to their fingers: Evidence for word-level verbal control of skilled action sequencing. International Meeting of the Psychonomic Society, Granada, Spain.

21 **Crump, M. J. C.**, & **Brosowsky, N. P. (2016). Context-specific proportion congruent effects for frequency unbiased items: Novel extensions and analyses of reproducibility. International Meeting of the Psychonomic Society, Granada, Spain.

22 *Behmer, L. P., & **Crump, M. J. C.** (2016). Knowing where to go when you get there: Dynamic construction of spatial knowledge during skilled action sequencing. International Meeting of the Psychonomic Society, Granada, Spain.

23 **Brosowsky, P. N., & **Crump, M. J. C.** (2016). Contextual control over stimulus-response sets. Psychonomics, Eastern Psychological Association, New York, NY.

24 *Behmer, L. P., & **Crump, M. J. C.** (2016). The dynamics of best laid action plans: Tracing response scheduling in skilled typing. Eastern Psychological Association, New York, NY.

25 Zumsteg, J., **Crump, M. J. C.**, Logan, G., Weikert, D., & Lee, D. (2016). The effect of carpal tunnel release surgery on post-operative typing function. 71st Annual Meeting of the American Society for Surgery of the Hand. Austin, TX.
26 *Behmer, L. P., & **Crump, M. J. C.** (2015). The dynamics of best laid action plans: Tracing response scheduling in skilled typing. Psychonomics. Chicago, IL.
27 **Brosowsky, P. N., & **Crump, M. J. C.** (2015). Contextual control of attentional sampling: Exploring the role of volition. Psychonomics, Chicago, IL.
28 **Crump, M. J. C.**, & *Behmer, L. (2015). Learning, memory and the control of serial order for action. International Workshop in Learning, Memory, and Attention, Unversidad de Granada, Granada, Spain.
29 **Brosowsky, N., & **Crump, M. J. C.** (2015). Evidence for generalization of context-specific control in a flanker task. Eastern Psychological Association, Philadelphia, PA.
30 *Behmer, L. P., & **Crump, M. J. C.** (2015). Typists develop sensitivity to the statistical frequencies of likely N-grams. Eastern Psychological Association, Philadelphia, PA.
31 **Crump, M. J. C.**, ***Blecher, M., & Jamieson, R. (2014). False memory for previously tested items: Investigating test-induced associative and repetition priming. Psychonomics, Long Beach, CA.
32 **Crump, M. J. C.** (2014). Learning and stimulus-driven cognitive control: Contextual cuing of explicit attention strategies. Eastern Psychological Association, Boston, MA.
33 **Mitchell, K. M., & **Crump, M. J. C.** (2013). Betting on your inner homunculus: Query theory in a dynamic gambling task. Eastern Psychological Association, New York City, NY.
34 **Crump, M. J. C.** (2013). Stimulus-driven control of explicitly induced attention filtering demands. Psychonomics, Toronto, ON, CA.
35 **Crump, M. J. C.**, Gureckis, T. M., & McDonnell, J. V. (2012). Can internet data be trusted? Validating mechanical turk for cognitive research. Psychonomics, Minneapolis, MN.
36 Jamieson, R. K., **Crump, M. J. C.**, & Hannah, S. D. (2012). A computer tool to simulate and teach associative learning. Society for Computers in Psychology, Minneapolis, MN.
37 **Crump, M. J. C.** (2012). Error detection in Skilled Typing: Ounce of Prevention, Pound of Cure. Eastern Psychological Association. Pittsburgh, PA.
38 Jamieson, R. K., Hannah, S. D., & **Crump, M. J. C.** (2012). An instance theory of associative learning. Eastern Psychological Association, Pittsburgh, PA.
39 Jamieson, R. K., Chubala, C. M., Hannah, S. D., & **Crump, M. J. C.** (2012). An instance-based account of retrospective revaluation. Psychonomics, Minneapolis, MI.
40 **Mitchell, K., & **Crump, M. J. C.** (2012). Poverty, attention and the power of the default option. Society for Judgment and Decision Making, Minneapolis, MI.
41 **Mitchell, K., & **Crump, M. J. C.** (2012). Continuous tracking of Contingency Changes. Eastern Psychological Association. Pittsburgh, PA.
42 **Crump, M. J. C.** (2011). Mapping the fretboard: Investigating spatial representations of note information in skilled guitarists. Canadian Society for Brain, Behavioural, and Cognitive Sciences, Winnipeg, MA, CA.
43 **Crump, M. J. C.** (2011). Hitting the wall in skilled typing: Speed/Accuracy tradeoffs and serial processing. Psychonomics. Seattle, WA.
44 Jamieson, R. K., Hannah, S. D., & **Crump, M. J. C.** (2011). A simple network model of associative learning (not memory). Society for Computers in Psychology. Seattle, WA.

Grants

Funded

Title	Agency	Amount	Period
Generative AI methods and applications for Psychology and the Cognitive Sciences	Brooklyn College Provost Teaching Innovations Grant 2025-26	$2,500	Spring 2026
Remembering and forgetting pictures: Testing a computational instance-based account	PSC-CUNY Traditional B	$6,000	2023-2024
Instances of cognition: An open-educational resource for introductory Cognitive Psychology	CUNY OER Initiative	$15,000	2021

Title	Agency	Amount	Period
Recruitment of cognitive control for action sequencing	PSC-CUNY Traditional B	$5,419.23	July 2017 - June 2018
Acquisition of hierarchical control in skilled action sequencing	National Science Foundation (#1353360)	$356,073	July 2014 - June 2017
Validating web-based technology for investigating strategic and stimulus-driven attentional control	PSC-CUNY Traditional A (#66163-00 44)	$3,478	Summer 2013
Chronometric investigations of explicit and implicit error detection processes in skilled typing	PSC-CUNY Traditional A (#65075-00 43)	$3,418	Summer 2012

Submitted

Title	Agency	Amount requested	Submitted
Answering questions with data: Teaching computational skills in introductory statistics to psychology undergraduates	National Science Foundation (PI)	$299,470	February 2021
Cognitive, competitive, and emergent control processes for action sequencing	National Science Foundation (PI)	$484,575	August 2017
Cognitive control for action sequencing	National Science Foundation (Co-PI)	$138,000	November 2016
Dynamics of cue-driven control over selection for attention and action	National Science Foundation (PI)	$411,790	August 2016
Designing and testing mediated physical spaces with human interaction sensing for increased multi-party goal satisfaction	National Science Foundation (Co-PI)	$980,172	January 2016
Contingency assessment ability and associative processing in normal, anxiety, orbitofrontal, and dorsolateral pre-frontal lesioned populations	National Institutes of Health (SCORE program) (PI)	$285,384 (direct costs)	January 2012
Unraveling the serial-order problem in skilled performance: Investigating dynamic online control and learning of sequential structure in skilled typing	CUNY Junior Faculty Research Awards in Science and Engineering (PI)	$50,000	November 2011
Control hierarchies in error detection and correction	PSC-CUNY (PI)	$5,000	February 2011

Professional Awards

- Doctoral Thesis Award, Certificate of Academic Excellence, Canadian Psychological Association (2008)
- Postgraduate Scholarship (PGS-D), Natural Sciences and Engineering Research Council of Canada (NSERC) (2005-2007)
- Ontario Graduate Scholarship (declined), Government of Ontario (2005)
- Ontario Graduate Scholarship, Government of Ontario (2003-2004)

Software and Other Creative Work

- **midiblender**: An R package for generative MIDI sequencing and analysis. https://www.crumplab.com/midiblender/
- **NSFrmarkdown**: An R Markdown template for NSF grants. https://github.com/CrumpLab/NSFrmarkdown
- **Semantic Librarian**: A website of cognitively inspired search engines for journal abstracts in the APA database, the CSBBCS conference database, the Society for Computation in Psychology database, and Psychonomics 2019. https://www.semanticlibrarian.com
- **conflictPower**: Monte Carlo power analysis for cognitive control designs. https://www.crumplab.com/conflictPower/
- **playjareyesores**: Functions for detecting textual overlap between documents. https://www.crumplab.com/playjareyesores/
- **jspsychr**: Write and run jsPsych experiments from RStudio. https://www.crumplab.com/jspsychr/

- **RsemanticLibrarian**: R functions for creating semantic librarians, a tool for vectorizing and visualizing semantic spaces from text. https://www.crumplab.com/RsemanticLibrarian/
- **vertical**: RStudio project template and workflow for sharing psychological research projects as a website. https://www.crumplab.com/vertical/

University, College and Community Service

Service to Brooklyn College

College committees

Spring 2021 - present	Curriculum Committee for the Interdisciplinary Undergraduate Major in Data Analytics

School and division committees

Fall 2024 - present	Faculty Council: Chair, Department of Psychology
Fall 2015 - May 2018	Faculty Council: Natural and Behavioral Sciences Representative

Department committees

Fall 2025 - present	Co-Chair, Honors and Awards Committee
Fall 2024 - present	Chair, Department of Psychology
2024	Department Self Study, Appendices working group
Fall 2022 - Spring 2023	Open Area Search Committee
2022 - present	Master of Science in Psychological Research program reboot committee
2021	Experimental Psychology Masters Comprehensive Exam Evaluator
Spring 2019	Co-chair, search committee for lecturer line
Spring 2019	Director of the Experimental Psychology Graduate Program
May 2018 - June 2021; May 2022 - present	Appointments Committee (elected)
2018, Spring 2019, Fall 2020, 2021, 2022, 2023	Experimental Psychology Masters Executive Committee
2018	Ad Hoc Independent Research Course Committee (chair)
February 2017	Doctoral Student Funding Committee
Fall 2014	Quantitative Job Search Committee
Fall 2014 - present	Department Website Author
Fall 2014 - Spring 2018	Library Technology Liaison
Summer 2014 - Fall 2014	Experimental Psychology Master's Program Curriculum Committee
Spring 2014	Psychology Undergraduate Program Assessments Ad Hoc Committee
2013, 2017, 2018, 2019	Experimental Masters Admissions Committee
Fall 2013	Behavioral Neuroscience Job Search Committee
Spring 2013	Master's Program Review Committee
Spring 2012 - present	Undergraduate Honors Committee
Fall 2011 - Spring 2013	Library Liaison

Student activities

Fall 2013 - 2015	Faculty liaison to the Undergraduate Psychology Club
Spring 2012, Fall 2012	Introductory lectures for visiting high school students

Other service

2011, 2012, 2013, 2014, 2016, 2017, 2021, 2022, 2023, 2024	Brooklyn College Honors Academy Reader
	Course pack author for PSYC 2530 Introduction to Cognitive Psychology
	Course pack author for PSYC 3530 Advanced Cognitive Psychology
	Website creator for the Cognitive and Comparative Psychology training area
	Brooklyn College Day coordinator
	Ad hoc Internet Satisfaction Committee

Service to the University and Graduate Center

University Service

Fall 2026 - present	Co-chair, Psychology Discipline Council, CUNY

Doctoral program committees

Fall 2024 - present	Executive Committee, Department of Psychology, Graduate Center of CUNY
2024	PSC-CUNY Grant Review Panel, Cycle 55 (reviewer)
Spring 2021	Psychology Quantitative Research Ad Hoc Committee

2018 - present	Cognition and Comparative Psychology Training Area Executive Committee
2017 - May 2018	Merger Committee to merge the Cognition, Language, and Development training area with the Animal Behavior and Comparative Psychology training area
January 2015 - May 2018	Cognition, Language, and Development Training Area Executive Committee
January 2015 - 2017	Training Area and Cluster Organization Committee
December 2014 - May 2018	Head of the Cognition, Language, and Development Training Area
Fall 2014, 2015, 2016, 2017, 2018	Cognition, Language, and Development Admissions Committee
Fall 2013	Cognition, Language, and Development Ad Hoc Curriculum Committee
Fall 2012	Cognition, Brain and Behavior Executive Committee

Service Off-Campus

Professional activities and memberships

2025	Guest Editor, Gordon D. Logan Festschrift (with D. W. Schneider and F. Verbruggen), The American Journal of Psychology, 138(2)
2020-2021	Diversity, Equity and Inclusion Committee, Society for Computation in Psychology (SCiP)
2019 - present	Consulting Editor, Psychonomic Bulletin & Review
2018 - present	Review Board, Journal of Experimental Psychology: Learning, Memory, and Cognition
Fall 2014 - present	Steering Committee, Society for Computation in Psychology (SCiP)
Fall 2012 - present	Review Board, Canadian Journal of Experimental Psychology
	Member, Eastern Psychological Association
	Member, Psychonomic Society

Reviewing

Grant reviewing

Spring 2017	Panel Reviewer, National Science Foundation (PAC division)
2016, 2017	External Grant Reviewer, National Science Foundation
February 2013	External Grant Reviewer, French National Research Agency

Ad hoc reviewer (2011 - present)

Acta Psychologica; Attention, Perception, and Psychophysics; Behavior Research Methods; Canadian Journal of Experimental Psychology; Cerebral Cortex; Cognition; Cognitive Psychology; Cognitive Science; Connection Science; Consciousness and Cognition; Current Directions in Psychological Science; Experimental Psychology; Frontiers in Cognition; Journal of Experimental Psychology: Applied; Journal of Experimental Psychology: Animal Learning and Cognition; Journal of Experimental Psychology: General; Journal of Experimental Psychology: Human Perception and Performance; Journal of Experimental Psychology: Learning, Memory, and Cognition; Perceptual and Motor Skills; Psicologica; Psychological Research; Psychonomic Bulletin & Review; Quarterly Journal of Experimental Psychology

Mentorship

Postdoctoral Associates

Lawrence Behmer	Summer 2014 - 2018

Doctoral Students

Drew Shives	2021-current; Cognitive and Comparative Psychology training area
Nicholaus Brosowsky	2014-2019; Cognition, Language, and Development training area
Kevonte Mitchell	2012-current; Brain, Cognition, and Behavior program
Felicia Valetta	2011-2014; Brain, Cognition, and Behavior program

Master's Students

Ashot Balayan	2 Independent Research, 1 Independent Reading, Master's thesis
Yi Sin Goh	1 Independent Research
Yashodra Khan	1 Independent Reading, 1 Independent Research
Kevonte Mitchell	1 Independent Reading, 1 Independent Research
Michelle Villavicencio	1 Independent Research, Master's thesis

Undergraduate Students

Michael Blecher	2 Independent Research
Ericka Boucard	1 Independent Research
Esthena Brutten	2 Independent Research, 1 Honors thesis, Scholars Program
Rebecca Colon	REU student (Research Experience for Undergraduates)
Elena Domingo	2 Independent Research, MARC (Maximizing Access to Research Careers)
Ece Ersoy	1 Laboratory Experience, Honors thesis, Macaulay Honors
Rochelle Hall	REU student
Jonathan Hagler	2 Independent Research
Wesley Huang	Tow Presidential Mentee (1st cohort), 2 Laboratory Experience
Patrick Ihejirika	1 MARC (Maximizing Access to Research Careers), Honors thesis, Goldwater Scholarship winner
Austin Kaplan	2 Independent Research, Honors thesis
Walter Lai	Volunteer research assistant
Shawn Lauzon	1 Independent Research
Myrodati Lyristis	1 Honors thesis, Scholars Program
Wing Jing Mak	1 Independent Research
Shifa Maqsood	Tow Presidential Mentee (2nd cohort), 1 Independent Research
Beni Mftidi	1 Independent Research
Kendra Scarlett Reynoso Morena	1 Independent Reading
Konstantin Nestoiter	1 Independent Research
Emily Paolillo	2 Independent Research, REU, Macaulay Honors Program
Florina Petcu	1 Independent Research, CUNY BA
Mark Rayev	1 Laboratory Experience, Honors thesis, Macaulay Honors
Ana Sanchez	REU student
Marissa Scotto	REU student
Chin Yen Tan	1 Independent Research
Niles Uy	1 Laboratory Experience, 2 Independent Research, 1 Honors thesis
Hao Zhu	1 Laboratory Experience, 1 Independent Research, CUNY BA

Dissertation Committees

2023	Committee member, dissertation defense (Janani Rajagopalan), Graduate Center of CUNY, New York, NY
2022	Committee member, dissertation defense (Kelly Cotton), Graduate Center of CUNY, New York, NY
2021	Committee member, dissertation defense (Melissa Horger), Graduate Center of CUNY, New York, NY
2019	Committee chair, dissertation defense (Nicholaus Brosowsky), Graduate Center of CUNY, New York, NY
2017	Committee member, dissertation defense (Daniel Meredith), Graduate Center of CUNY, New York, NY
2016	External reader, dissertation defense (Rachel Searston), School of Psychology, The University of Queensland, Australia
2016	Committee member, dissertation defense (Natasha Nadler), Graduate Center of CUNY, New York, NY
2015	Committee member, dissertation defense (Denise Pergolizzi), Graduate Center of CUNY, New York, NY
2013	Committee member, dissertation defense (Justin Ostrofsky), Graduate Center of CUNY, New York, NY
2013	External reader, dissertation defense (John McDonnell), Department of Psychology, New York University, New York, NY
2013	External reader, dissertation defense (Maryem Torres-Quesada), Department of Psychology, University of Granada, Granada, Spain
2012	External reader, dissertation defense (Elena Cañadas), Department of Psychology, University of Granada, Granada, Spain

Teaching Experience

OER refers to open educational resources authored by M. J. C. Crump. These are Creative Commons licensed resources that are free for students to use, and for others to copy, remix, and reuse.

Undergraduate Courses

Course	Title	Notes
PSYC 2530	Introduction to Cognitive Psychology https://www.crumplab.com/cognition/	In-person, async, sync, large sections, OER
PSYC 3400	Statistical Methods in Psychological Research https://www.crumplab.com/statistics/	Large sections, OER
PSYC 3450W	Experimental Psychology https://www.crumplab.com/ResearchMethods/	Large sections, OER
PSYC 3470	Data Analytics and Data Visualization in the Social and Behavioral Sciences	OER
PSYC 3530	Advanced Cognitive Psychology	
PSYC 2001	Laboratory Experiences	Student mentoring
PSYC 5001	Independent Research I	Student mentoring
PSYC 5002	Independent Research II	Student mentoring
PSYC 5003	Independent Research III	Student mentoring
MCHC 1002	Seminar 2: People of New York City	Macaulay Honors College, outside the Psychology Department

Master's Courses

Course	Title	Notes
PSYC 7765G	Statistical Methods Application I https://www.crumplab.com/rstatsforpsych/	OER
PSYC 7766G	Statistical Methods Application II https://www.crumplab.com/rstatsmethods/	OER
PSYC 7709G	Special Topics: Using R for Reproducible Psychological Research https://www.crumplab.com/psyc7709_2019/	OER
PSYC 7792G	Independent Reading	Student mentoring
PSYC 7795G	Independent Psychological Research I	Student mentoring
PSYC 7796G	Independent Psychological Research II	Student mentoring
PSYC 7781G	Research in Psychology	Master's theses

Doctoral Courses

Course	Title	Notes
PSYC 73800	Cognitive Psychology	Course organizer
PSYC 80200	Independent Study	Student mentoring
PSYC 80102	Seminar in Special Topics: CCP Colloquium	
PSYC 80103	Special Topics: Computer Programming for Experimental Psychology https://www.crumplab.com/programmingforpsych/	OER
PSYC 80103	Special Topics: Cognitive Technologies: From Theory and Data to Application	OER
PSYC 80103	Special Topics: Memory: Foundations and Current Issues	Co-taught with E. Chua
PSYC 80103	Special Topics Seminar: Learning and Attention	Co-taught with A. Delamater
PSYC 72400	Career Development	Co-taught with E. Chua
PSYC 90000	Dissertation Supervision	

Teaching Certificates

- Effective Practice Framework certificate in effective college instruction, Association of College and University Educators and the American Council on Education (2020-2021). A 26-week course.